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April 27, 1973
SUPERSEDING
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March 16, 1967

FEDERAL SPECIFICATION

TUBE FITTINGS, COPPER AND COPPER ALLOY, WROUGHT OR CAST, SOLDER ENDS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers solder-end tube fittings for use with copper water tubes.

1.2 Classification.

1.2.1 Types. Fittings furnished under this specification shall be of the following types and joining ends, as specified (see 6.2):

Type I - Wrought copper or copper-tin alloy.
Type II - Cast copper-zinc-tin-lead alloy.

1.2.2 Kinds, sizes, and ends. The various kinds and sizes of fittings with all solder-end connections, or with solder- and threaded-end connections, shall be in accordance with table I (see 6.2).

TABLE I. Kind, size, and end designation of fittings

Fitting	Standard size		Fitting symbol 1/	
	inch	mm	cast	wrought
Adapter	1/2	12.7	---	C to M
	1/2	12.7	---	C to F
	1/2	12.7	---	FTG to F
	3/4	19.0	---	C to M
	3/4	19.0	---	C to F
	3/4	19.0	---	FTG to F
	1/2 X 3/4	12.7 X 19.0	---	C to M
Cap	1/2	12.7	---	C
	3/4	19.0	---	C
Coupling	1/2	12.7	---	C to C
	3/4	19.0	---	C to C
	1/2 X 3/4	12.7 X 19.0	---	C to C
Elbow 90°	1/2	12.7	C to F	C to C
	3/4	19.0	C to F	C to C
	1/2 X 3/4	12.7 X 19.0	C to F	C to C
	1/2 X 3/8	12.7 X 9.52	C to F	----
	3/4 X 1/2	19.0 X 12.7	C to F	----
Elbow 90° street	1/2	12.7	---	C to FTG
	3/4	19.0	---	C to FTG
Elbow 45°	1/2	12.7	---	C to C
	3/4	19.0	---	C to C
Elbow 45° street	1/2	12.7	---	C to FTG
	3/4	19.0	---	C to FTG
Plug	1/2	12.7	---	FTG
	3/4	19.0	---	FTG
Reducer	3/4 X 1/2	19.0 X 12.7	---	FTG to C
	1 1/4 X 1	31.7 X 25.4	---	FTG to C

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TABLE I. Kind, size, and end designation of fittings (con.)

Fitting	Standard size		Fitting symbol 1/	
	inch	mm	cast	wrought
Tee	1/2	12.7	---	C to C to C
	3/4	19.0	---	C to C to C
	1/2 X 1/2 X 3/4	12.7 X 12.7 X 19.0	---	C to C to C
	3/4 X 1/2 X 1/2	19.0 X 12.7 X 12.7	---	C to C to C
	3/4 X 1/2 X 3/4	19.0 X 12.7 X 19.0	---	C to C to C
	3/4 X 3/4 X 1/2	19.0 X 19.0 X 12.7	---	C to C to C

1/ Symbols for designation of fitting ends:

C - Solder-joint fitting end (female) machined or made to receive copper tube diameter.
FTG - Solder-joint fitting end (male) machined or made to copper tube diameter.
M - External USA American Standard Taper Pipe Thread (male).
F - Internal USA American Standard Taper Pipe Thread (female).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-V-3 - Vales, Fittings and Flanges, Packing of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American National Standards Institute (ANSI) Standards:

B16.18 - Cast Bronze Solder-Joint Pressure Fittings.
B16.22 - Wrought Copper and Bronze Solder-Joint Pressure Fittings.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Type I, wrought. Fittings shall conform to the requirements of ANSI Std B16.22.

3.2 Type II, cast. Fittings shall conform to the requirements of ANSI Std. B16.18.

3.3 Workmanship. Fittings shall be clean both internally and externally with roughness removed. Solder or threaded ends shall be of full depth. Metal on parting line of cast fittings shall be ground flush. Cast fittings shall be free of cracks and in-line surface porosity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that the supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all of each type and fitting offered for inspection at the same time.

4.2.2 Visual, dimensional, and workmanship. Sampling for these characteristics shall be in accordance with level S-3 of MIL-STD-105.

4.2.2.1 Samples for tests. Sampling for tests shall be in accordance with level S-2 of MIL-STD-105.

4.3 Examination.

4.3.1 End item. Fittings shall be examined for defects listed in table II. The Acceptable Quality Levels (AQL's) shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

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TABLE II. Classification of defects, end item

Defects	Major	Minor
Type and fitting not as specified.	X	
Dimensions not within tolerances.	X	
Solder or threaded end ovalty not within tolerances.	X	
Length or depth of soldered ends less than the minimum length or depth in table I of ANSI Std. B16.18 for cast fittings, or in table I of ANSI Std. B16.22 for wrought fittings.	X	
Fittings not smooth or clean.		X
Cracks or in-line surface porosity in cast fittings.	X	
Parting line not flush on cast fittings.		X

4.3.2 Inspection of preparation for delivery. An inspection shall be made to determine that the preservation, packaging, packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with Table III. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sample shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers in the end item inspection lot. The inspection level shall be II and the AQL shall be 4.0 defects per hundred units.

TABLE III. Classification of preparation for delivery defects

Examine	Defects
Marking (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of component such as incomplete closure of container flaps, loose strapping, inadequate stapling, or distortion of container.

4.4 Test methods. Fittings shall be tested at an air pressure of 60 psig minimum while under water, or to a hydrostatic pressure of 250 psig minimum. The AQL for tests shall be 4.0 percent defective.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. The fittings furnished in quantities as specified (see 6.2) shall be preserved, packaged and packed in accordance with MIL-V-3. The level of preservation and packaging shall be in A or C and the level of packing shall be A, B, or C as specified (see 6.2).

5.2 Marking.

5.2.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military activities. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Wrought and cast fittings are intended for use with copper water tubes.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type (see 1.2.1).
(c) Kind of fitting (see 1.2.2).
(d) Size of lot, if different from 4.2.1.
(e) Selection of the applicable levels of packaging and packing required (see 5.1 and 6.3).

6.3 For civil agency procurements. When level B packaging is required for civil agency procurements, the level A requirements in 5.1 shall apply.

6.4 The fittings procured under this specification, which are centrally inventory managed, shall be limited to the varieties set forth in Fed. Std. No. 263. The specification may be used for the procurement of additional varieties which are not centrally procured or are approved in accordance with existing regulations.

6.5 Pressure rating. Fittings may be used for the same pressure-temperature ratings as straight seamless type L annealed copper water tube.

MILITARY CUSTODIANS:

Army - ME
Air Force - 82

User Activities:

Army - CE, GL, WC
Navy - MC, CG

Military Coordinating Activity:

Air Force - 82

Preparing Activity:

GSA-FSS

Civil Agency Coordinating Activities:

COMMERCE-NBS
GSA-FSS
HEW-FEC
USDA-AFS

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